MATHEMATICS

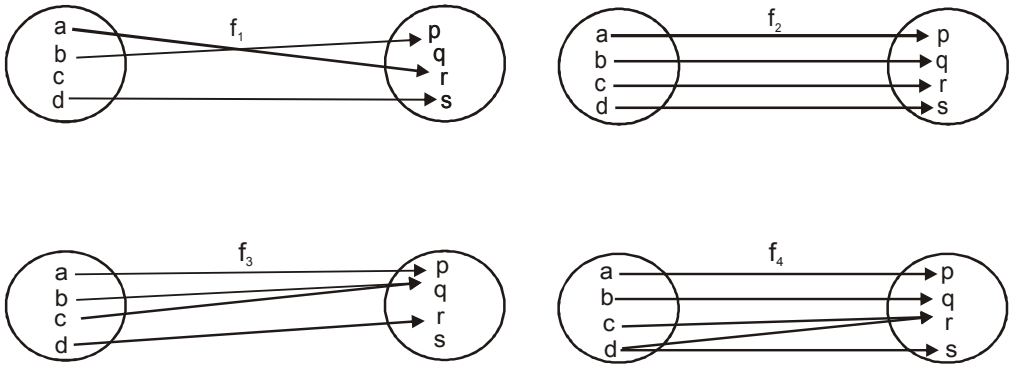
FUNCTION
THEORY SHEET
ENGLISH

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1.1 DEFINITION OF FUNCTION

Function is a relation from set A to set B such that each of every element of set A is related to exactly one element of set B.

Let P = {a, b, c, d} and Q = {p, q, r, s} be two sets and let f_1, f_2, f_3 and f_4 be rules associating elements of P to elements of Q as shown in the following figures.



Let P and Q be two non–empty sets. Then a function 'f ' from set P to set Q is a rule which associates elements of set P to elements of set Q such that an element of set P are associated to a unique element in set Q and all elements of set P are associated to element in set Q.

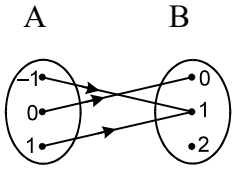
If 'f' is a function from a set P to set Q, then we write $f : P \to Q$ or $P \xrightarrow{f} Q$. which is read as f is a function from P to Q or f maps P to Q.

Terms such as "map" (or mapping), "correspondence" are used as synonyms for function.

For example, let $A \equiv \{-1, 0, 1\}$ and $B \equiv \{0, 1, 2\}$.

Now, " $f: A \to B$ defined by $f(x) = x^2$ " is the function such that $f \equiv \{(-1, 1), (0, 0), (1, 1)\}$

f can also be shown diagramatically by following mapping.



If an element $a \in A$ is associated with an element $b \in B$, then

1. (i) b is called 'the *f* image of a' or
   (ii) b is called 'image of a under *f*
   (ii) b is called the value of the function *f* at a.
2. (i) a is called the pre-image of b or
   (ii) a is argument of b under the function *f*.

$b = f(a)$ or $f : a \to b$ or $f : (a, b)$

1.2 DOMAIN, CO-DOMAIN AND RANGE OF A FUNCTION

Let $f : A \to B$, then the set A is known as the domain of f & the set B is known as co-domain of f. The set of f images of all the elements of A is known as the range of f.

Thus : Domain of $f = \{a \mid a \in A, (a, f(a)) \in f\}$

Range of $f = \{f(a) \mid a \in A, f(a) \in B\}$

Note :

It should be noted that range is a subset of co-domain.

1.3 TYPES OF FUNCTION

(i) Uniform function :

$f(x) = x^2$

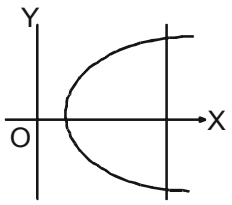
(ii) Non - uniform function :

$$f(x) = \begin{cases} x+1 & -1 \le x < 4 \\ -x & 4 \le x < 7 \end{cases}$$

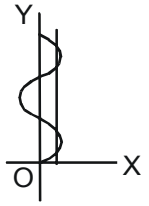
$$f(x) = \begin{cases} x^2 & x \ge 0 \\ -x-1 & x < 0 \end{cases}$$

1.4 GRAPHICAL REPRESENTATION OF FUNCTION

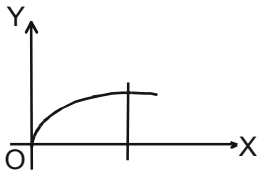
Vertical Line Test : If we are given a graph of the relation then we can check whether the given relation is function or not . If it is possible to draw a vertical line which cuts the given curve at more than one point then given relation is not a function and when this vertical line means line parallel to Y - axis cuts the curve at only one point then it is a function.



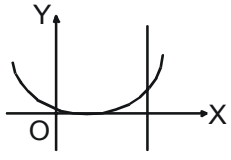
(i)



(ii)



(iii)



(iv)

fig. (iii) and (iv) represents a function.

1.5 ALGEBRA OF FUNCTION

If f & g are real valued functions of x with domain set A, B respectively,

(a) $(f \pm g)(x) = f(x) \pm g(x)$ domain in each case is $A \cap B$

(b) $(f.g)(x) = f(x).g(x)$ domain is $A \cap B$

(c) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ domain $A \cap B - \{x|g(x) = 0\}$

Note :

(i) If only the rule of function is given then the domain of the function is the set of those real numbers, where function is defined.

(ii) For a continuous function, the interval from minimum to maximum value of a function gives the range

(iii) We will not simplify a function to compute domain.

$f(x) = \frac{x^2 - 4}{x - 2}$ domain $= R - \{2\}$

$g(x) = x + 2$; domain $= R$

2.1 TYPES OF STANDARD FUNCTION

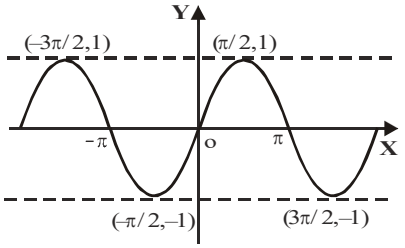
2.1.1 Trigonometric functions

(i) Sine function

$f(x) = \sin x$

Domain : R

Range : [–1, 1], period 2π

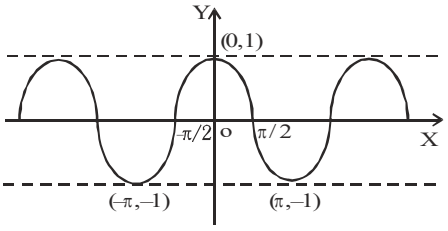


(ii) Cosine function

$f(x) = \cos x$

Domain : R

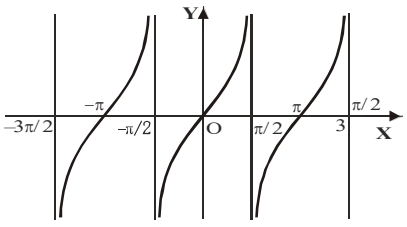
Range : [–1, 1], period 2π



(iii) Tangent function

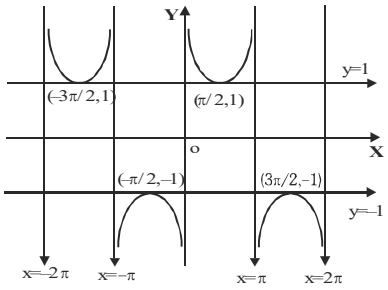
$f(x) = \tan x$

Domain : $R - \left\{x \mid x = \frac{(2n+1)\pi}{2}, n \in I\right\}$



Range : R , period π

(iv) Cosecant function

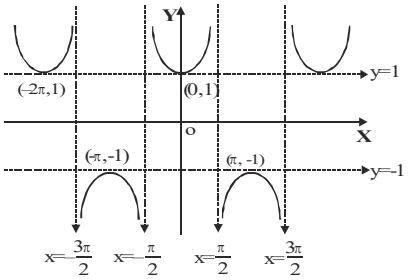


$f(x) = \text{cosec } x$

Domain : $R - \{x | x = n\pi, n \in I\}$

Range : $R - (-1, 1)$, period 2π

(v) Secant function

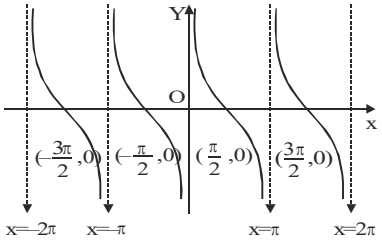


$f(x) = \sec x$

Domain : $R - \{x | x = (2n + 1)\ \pi/2 : n \in I\}$

Range : $R - (-1, 1)$, period 2π

(vi) Cotangent function



$f(x) = \cot x$

Domain : $R - \{x | x = n\pi, n \in I\}$

Range : R, period π

Find Domain of following functions

Q.1 $f(x) = \cos\left(\frac{1}{x}\right)$

Ans. $x \in R - \{0\}$

Q.2 $f(x) = \tan(2x)$

Ans. $x \in R - \left\{(2n+1)\frac{\pi}{4}\right\}$

Q.3 $f(x) = \cot\left(2x - \frac{\pi}{3}\right)$

Ans. $R - \left\{\frac{n\pi}{2} + \frac{\pi}{6}\right\}$

Q.4 $f(x) = \frac{1}{\text{cosec } x}$

Ans. $R - \{n\pi\}$

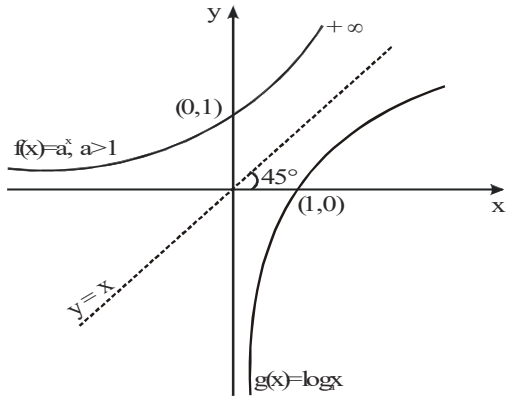
Q.5 $f(x) = \dfrac{1}{\tan x}$

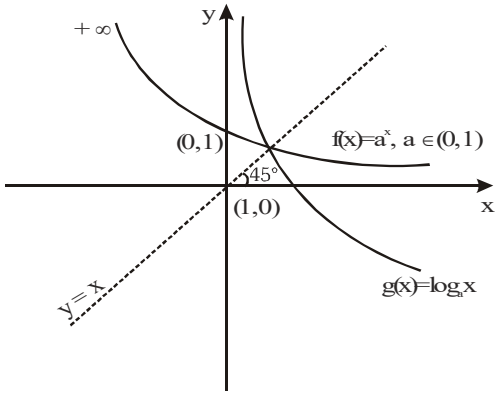
Ans. $R - \left\{\dfrac{n\pi}{2}\right\}$

2.1.2 Exponential Function

$f(x) = a^x (a > 0),\ a \neq 1,\ x \in R$

Domain of a^x is R **Range** R^+





Find Domain of following functions

Q.1 $f(x) = \sqrt{3^x - 9}$

Ans. $x \geq 2$

Q.2 $f(x) = \sqrt{8 - 2^x}$

Ans. $x \leq 3$

Q.3 $f(x) = \sqrt{7 - 2^x}$

Ans. $x \leq \log_2(7)$

Q.4 $f(x) = \sqrt{2^x + 1}$

Ans. $x \in R$

Q.5 $f(x) = \dfrac{1}{3^x - 9}$

Ans. $R - \{2\}$

Q.6 $f(x) = e^{\tan x}$

Ans. $R - \left\{(2n+1)\dfrac{\pi}{2}\right\}$

Q.7 $f(x) = x^x$

Ans. $x > 0$

Note : $(p(x))^{q(x)} = e^{q(x) \ln p(x)}$

2.1.3 Logarithmic Function

$f(x) = \log_a x$.

Domain of $\log_a x$ is R^+ **Range** R

Find Domain of following functions

Q.1 $f(x) = \sqrt{2 - \log_3 x}$

Ans. $0 < x \le 9$

Q.2 $f(x) = \log_2 (9 - 3x)$

Ans. $x < 3$

Q.3 $f(x) = e^{\ln x}$

Ans. $x > 0$

Q.4 $f(x) = \dfrac{1}{\ln x}$

Ans. $(0, \infty) - \{1\}$

Q.5 $f(x) = \log_2 x + \sqrt{9 - x}$

Ans. $x \in (0, 9]$

Sol. $x > 0$ & $9 - x \ge 0$

$x \le 9$

$x \in (0, 9]$

Q.6 $f(x) = \log_x (3 - x)$

Ans. $(0,1) \cup (1,3)$

Sol. $x > 0$ & $x \ne 1$ & $3 - x > 0$

$x < 3$

$x \in (0,1) \cup (1,3)$

2.1.4 Polynomial function

Standard Form

$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + ... + a_{n-1} x + a_n$

where n is a non negative integer and $a_0, a_1, a_2, ...a_n$ are real numbers and $a_0 \ne 0$,

Important Points

1. All powers of x are non-negative integers

2. Highest power of x is called degree of polynomial (i.e. n).

3. Coefficient of highest power of x is called leading coefficient i.e. a_0

4. $a_0 \neq 0$

5. a_n = constant term/ absolute term/ term independent of x

6. n = 1 linear $f(x) = ax + b$

   n = 2 quadratic $f(x) = ax^2 + bx + c$

   n = 3 cubic $f(x) = ax^3 + bx^2 + cx + d$

7. A polynomial of degree one with no constant term is called an odd linear function.i.e. $f(x) = ax, a \neq 0$

8. Domain of a polynomial function is R

9. Range of odd degree polynomial is R whereas range of an even degree polynomial is never R.

Eg. $f(x) = x^5 + x^2 + 3$

$$= x^5\left(1 + \frac{1}{x^3} + \frac{3}{x^5}\right)$$

$x \to \infty \Rightarrow y \to \infty; x \to -\infty \Rightarrow y \to -\infty$

Eg. $f(x) = x^4 - 3x + 2$

$$= x^4\left(1 - \frac{3}{x^3} + \frac{2}{x^4}\right)$$

$x \to \infty \Rightarrow y \to \infty; x \to -\infty \Rightarrow y \to \infty$

Note:

(i) Odd degree polynomial will have atleast one zero.

(ii) There are two polynomial functions, satisfying the relation;
$f(x).\ f(1/x) = f(x) + f(1/x)$. They are $f(x) = x^n + 1$ & $f(x) = 1 - x^n$, where n is a positive integer.

2.1.5 Rational function :

A rational function is a function of the form $y = f(x) = \frac{g(x)}{h(x)}$,
where g(x) & h(x) are polynomials & $h(x) \neq 0$,

Domain : $R - \{x \mid h(x)=0\}$

2.1.6 Algebraic function :

A function 'f' is called an algebraic function if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking radicals) starting with polynomials.

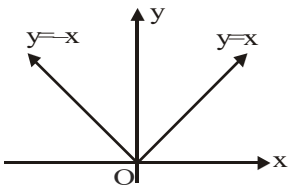
Examples : $f(x) = \sqrt{x^2+1}$; $g(x) = \dfrac{x^4 - 16x^2}{x+\sqrt{x}} + (x-2)\sqrt[3]{x+1}$

Note that all polynomial functions are Algebraic but the converse in not true. A functions that is not algebraic is called **TRANSCEDENTAL** function.

2.1.7 Modulus Function (Absolute value function) :

The absolute value (or modulus) of a real number x (written |x|) is a non negative real number that satisfies the conditions.

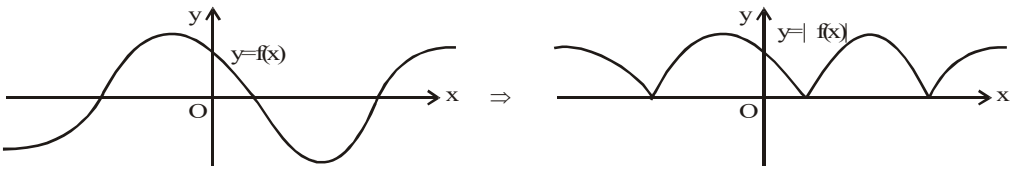
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



Domain : R
Range : $[0, \infty)$

Plot the graph of y = |f(x)| from the graph of y = f(x)

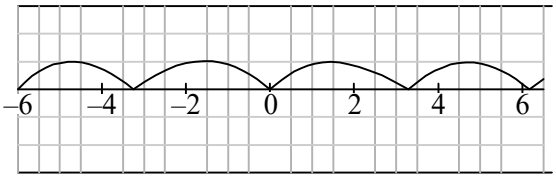
|f(x)| = f(x) if f(x) ≥ 0 and |f(x)| = –f(x) if f(x) < 0. It means that the graph of f(x) and |f(x)| would coincide if f(x) ≥ 0 and for the portions where f(x) < 0 graph of |f(x)| would be image of y = f(x) in x-axis.



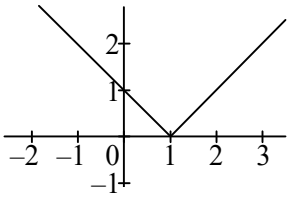
Q.1 Plot graph of y = |sin x|

→ first plot graph of y = sin x

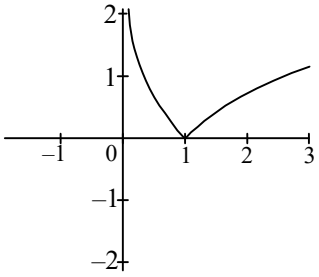
→ Remove the portion of graph line below x - axis and take its image w.r.t. x - axis.



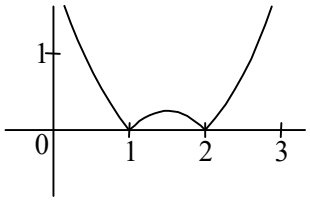
Q.2 y = |x – 1|



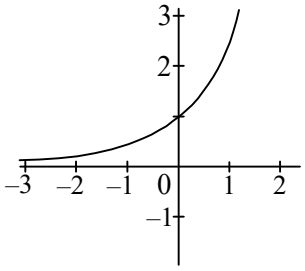
Q.3 y = |ln x|



Q.4 $y = |(x-1)(x-2)|$



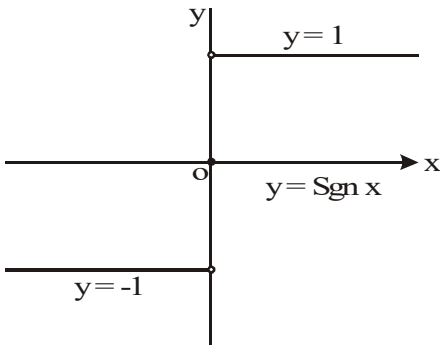
Q.5 $y = |e^x|$



2.1.8 Signum function

Signum function $y = \text{sgn}(x)$ is defined as follows

$$y = \begin{cases} \frac{|x|}{x}, x \neq 0 \\ 0, x = 0 \end{cases} = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$



Domain : R **Range** : $\{-1, 0, 1\}$

Properties :

1. $\text{sgn}(x) = 0 \Rightarrow x = 0$

   $\text{sgn}(x) = 1 \Rightarrow x > 0$

   $\text{sgn}(x) = -1 \Rightarrow x < 0$

2. $x\,\text{sgn}(x) = |x|$

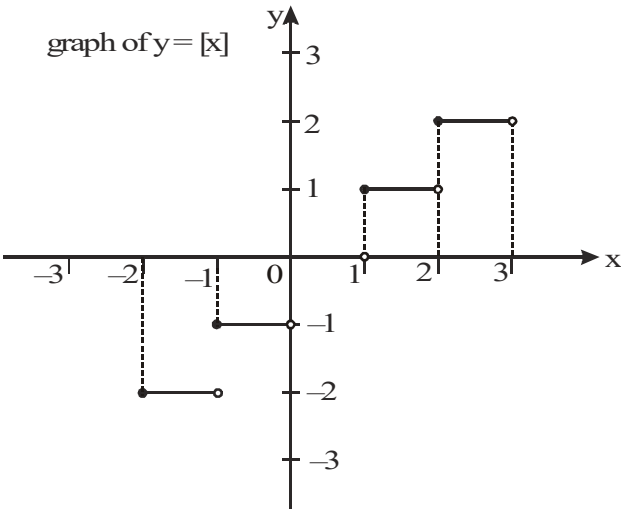
   $|x|\,\text{sgn}(x) = x$

3. $\text{sgn}(\text{sgn}(x)) = \text{sgn}(x)$

2.1.9 Greatest integer or step up function :

The function $y = f(x) = [x]$ is called the greatest integer function where $[x]$ denotes the greatest integer less than or equal to x. Note that for :

x	[x]
[–2,–1)	–2
[–1,0)	–1
[0,1)	0
[1,2)	1



Domain : R

Range : I

Properties of greatest integer function :

(i) $[x] \le x < [x] + 1$ and $x - 1 < [x] \le x$, $0 \le x - [x] < 1$

(ii) $[x + m] = [x] + m$ if m is an integer.

(iii) $[x] + [-x] = \begin{cases} 0, & x \in I \\ -1, & x \notin I \end{cases}$

(iv) $[x] + [y] \le [x + y] \le [x] + [y] + 1$

Define

Q.1 $f(x) = [x] \quad x \in [1.2, 1.8]$

Ans. $f(x) = 1$

Q2 $f(x) = [x] \quad x \in [1.2, 2.8]$

Ans. $f(x) = 1$ when $x \in [1.2, 2)$ and $f(x) = 2$ if $x \in [2, 2.8]$

Q.3 $f(x) = [2x] \quad x \in [1.2, 1.8]$

Ans. $f(x) = 2$ if $x \in [1.2, 1.5)$ and $f(x) = 3$ if $x \in [1.5, 1.8)$

Q4 $f(x) = [\sin^2 x]\ x \in [0, \pi]$

Ans. $f(x) = 0$ if $x \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$ and $f(x) = 1$ if $x \in \left\{\frac{\pi}{2}\right\}$

Q.5 $f(x) = [x^2] \quad x \in [0, 2]$

Ans. $f(x) = \begin{cases} 0 & \text{if} \quad 0 \le x < 1 \\ 1 & \text{if} \quad 1 \le x < \sqrt{2} \\ 2 & \text{if} \quad \sqrt{2} \le x < \sqrt{3} \\ 3 & \text{if} \quad \sqrt{3} \le x < 2 \\ 4 & \text{if} \quad 2 = x \end{cases}$

Q.6 $f(x) = [\sqrt{x}\ ]\ x \in [0, 8]$

Ans. $f(x) = \begin{cases} 0 & \text{if} \quad 0 \le x < 1 \\ 1 & \text{if} \quad 1 \le x < 4 \\ 2 & \text{if} \quad 4 \le x \le 8 \end{cases}$

Solve

1. $[x] = 3$

Ans $x \in [3, 4)$

2. $[x] = -5$

Ans $x \in [-5, -4)$

3. $[x] = \frac{3}{2}$

Ans $x \in \phi$

4. $[x] \ge 3$

Ans $x \ge 3$

$[\ [x] \ge I \Rightarrow x \ge I;\ [x] \le I \Rightarrow x < I + 1\ ]$

5. $[x] \ge -5$

Ans $x \ge -5$

6. $[x] \le 7$

Ans $x < 8$

7. $[x] \le -4$

Ans $x < -3$

8. $[x] < 5$

Ans $[x] \le 4 \Rightarrow x < 5$

9. $[x] < 7.6$

Ans $[x] \le 7 \Rightarrow x < 8$

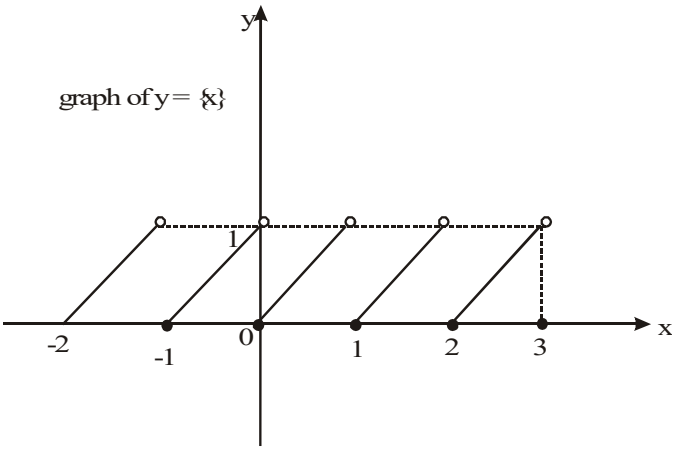
10. $[x] > 5.6$

Ans $[x] \ge 6 \Rightarrow x \ge 6$

2.1.10 Fractional part function :

It is defined as : $g(x) = \{x\} = x - [x]$ e.g.

the fractional part of the number 2.1 is $2.1 - 2 = 0.1$ and the fractional part of -3.7 is 0.3 graph of this function is as shown.



x	{x}
[–2,–1)	x+2
[–1,0)	x+1
[0,1)	x
[1,2)	x–1

Domain : R **Range :** [0,1)

Properties of Fractional part function :

(i) $0 \le \{x\} < 1$ (ii) $\{[x]\} = [\{x\}] = 0$

(iii) $\{\{x\}\} = \{x\}$ (iv) $\{x+m\} = \{x\}, m \in I$

(v) $\{x\} + \{-x\} = \begin{cases} 1, & x \notin I \\ 0, & x \in I \end{cases}$

Other Important Points

1. $\sqrt[n]{a}$ = n^{th} root of $a = a^{\frac{1}{n}}$

$n \in N ; n \ge 2$

2. $a^{\frac{p}{q}}$: q = even ⟵ $a \geq 0$; q = odd ⟶ $a \in R$

3. ${}^{n}C_{r}$, ${}^{n}P_{r}$

$n \geq r$; $n \in N$, $r \in W$

4. max. (2, 3) = 3; min. (2, 3, 4) = 2; max. (2, 2) = 2; min. (2, 2) = 2

$f(x) = \max.\ (p(x), q(x))$

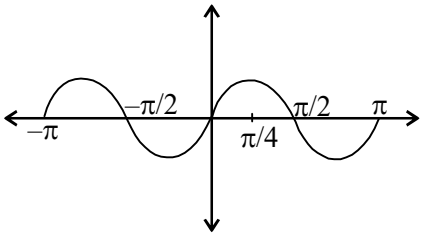
D_f = Intersection of D_p & D_q

Example Find $f(x) = \max\ \{1 + x, 1 - x, 2\}$.

Plot graph of y = f(ax) from the graph of y = f(x)

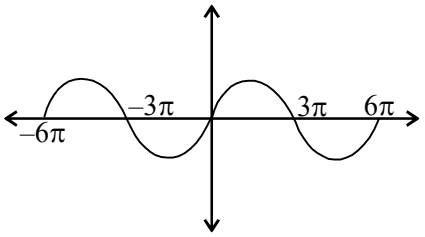
Q.1 $y = \sin(2x)$

Sol.



Q.2 $y = \sin\left(\frac{x}{3}\right)$

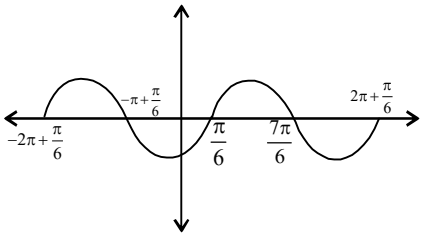
Sol.



Plot graph of y = f(x + a), a ∈ R from the graph of y = f(x)

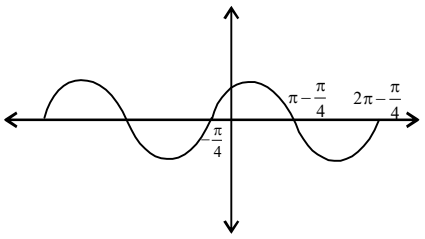
Q.1 $y = \sin\left(x - \frac{\pi}{6}\right)$

Sol.



Q.2 $y = \sin\left(x + \frac{\pi}{4}\right)$

Sol.



2.2 DOMAIN BASED QUESTIONS

Find domain

1. $f(x)=\sqrt{\dfrac{x-3}{x+2}}$.

Ans. $x\in(-\infty,-2)\cup[3,\infty)$

Sol. $\dfrac{x-3}{x+2}\geq 0$

$x\in(-\infty,-2)\cup[3,\infty)$

2. $f(x)=\sqrt{\dfrac{1-5^x}{7^{-x}-7}}$.

Ans. $x\in(-\infty,-1)\cup[0,\infty)$

Sol. $\left(\dfrac{5^x-1}{7^{x+1}-1}\right)\geq 0$

$x\in(-\infty,-1)\cup[0,\infty)$

3. $f(x)=\sqrt{x-|x|}$

Ans. $x\in[0,\infty)$

Sol. $x\geq|x|$

$x\in[0,\infty)$

4. $f(x)=\dfrac{1}{\sqrt{|x|-x}}$

Ans. $x\in(-\infty,0)$

Sol. $|x|>x$

$x\in(-\infty,0)$

5. $f(x)=\dfrac{1}{\sqrt{x-|x|}}$

Ans. $x\in\phi$

Sol. $x>|x|$

$x\in\phi$

6. $f(x)=\sqrt{\log_2\left(\dfrac{5x-x^2}{4}\right)}$

Ans. $x\in[1,4]$

Sol. $\log_2\left(\frac{5x-x^2}{4}\right)\geq 0$ & $5x-x^2>0$

$\frac{5x-x^2}{4}\geq 1$ $x(x-5)<0$

$5x-x^2\geq 4$ $x\in(0,5)$

$x^2-5x+4\leq 0$

$(x-1)(x-4)\leq 0$

$x\in[1,4]$

7. $f(x)=\log_2(6+35x-6x^2)+\sqrt{\cos(\sin x)}$

Ans. $x\in\left(-\frac{1}{6},6\right)$

Sol. $6+35x-6x^2>0$ & $\cos(\sin x)\geq 0$

$6x^2-36x+x-6<0$ $\cos[-1,1]\geq 0$

$(6x+1)(x-6)<0$ $x\in R$

$x\in\left(-\frac{1}{6},6\right)$

8. $f(x)=\sqrt{\frac{x-2}{x+2}}+\sqrt{\frac{1-x}{1+x}}$.

Ans. $x\in\phi$

Sol. $\frac{x-2}{x+2}\geq 0$ & $\frac{x-1}{x+1}\leq 0$

$x\in(-\infty,-2)\cup[2,\infty)$ $x\in(-1,1]$

9. $f(x)=\sqrt{(x^2-3x-10)\ln^2(x-3)}$

Ans. $x\in[5,\infty)\cup\{4\}$

Sol. $x^2-3x-10\geq 0$ & $x-3>0$ or $\ell n(x-3)=0$

$(x-5)(x+2)\geq 0$ & $x>3$ $x=4$

$x\in(-\infty,-2]\cup[5,\infty)$

$x\in[5,\infty)$

$x\in[5,\infty)\cup\{4\}$

10. $f(x)=\ln(\sqrt{x+2}-x)$

Ans. $x\in[-2,2)$

Sol. $x+2\geq 0\Rightarrow x\in[-2,\infty)$

$\sqrt{x+2}>x$

Case-I : $x\in[-2,0)$ **Case-II :** $x\in[0,\infty)$

+ve > – ve squaring

$x\in[-2,0)$ $x+2>x^2$

$x^2-x-2<0$

$(x-2)(x+1)<0$

$x\in[0,2)$

$x \in [-2, 2)$

11. $f(x) = \sqrt{\sin x} + \sqrt{16 - x^2}$.

Ans $[-4, -\pi] \cup [0, \pi]$

Sol $\sin x \geq 0$ and $16 - x^2 \geq 0 \quad \Rightarrow \quad 2n\pi \leq x \leq (2n + 1)\pi$ and $-4 \leq x \leq 4$

$\therefore$ Domain is $[-4, -\pi] \cup [0, \pi]$

12. $f(x) = \sqrt{\sin 2x} + \sqrt{9 - x^2}$.

Ans. $x \in \left[-3, -\frac{\pi}{2}\right] \cup \left[0, \frac{\pi}{2}\right]$

Sol. $x^2 - 9 \leq 0$

$x \in [-3, 3] \Rightarrow 2x \in [-6, 6]$

$\sin 2x \geq 0$

$2x \in [-6, -\pi] \cup [0, \pi]$

$x \in \left[-3, -\frac{\pi}{2}\right] \cup \left[0, \frac{\pi}{2}\right]$

13. $f(x) = \log_{[x]} (6 + 35x - 6x^2) + \frac{1}{[x]}$.

Ans. $x \in [2, 6)$

Sol. $6 + 35x - 6x^2 > 0$ & $[x] = 2,3,4,.....$

$x \in \left(-\frac{1}{6}, 6\right) \qquad x \in [2, \infty)$

$x \in [2, 6)$

14. $f(x) = \sqrt{[\{\ln x\}]} + \sqrt{9 - x^2}$.

Ans. $x \in (0, 3]$

Sol. $x^2 - 9 \leq 0 \quad$ & $\quad x > 0$

$x \in [-3, 3]$

$x \in (0, 3]$

15. $f(x) = \sqrt{\ln x} + \sqrt{9 - x^2}$.

Ans. $x \in [1, 3]$

Sol. $\ln x \geq 0 \quad$ & $\quad x > 0 \quad$ & $\quad 9 - x^2 \geq 0$

$x \geq 1 \qquad\qquad\qquad\qquad x \in [-3, 3]$

$x \in [1, 3]$

16. $f(x) = \sqrt{[x]^2 - 5} + \sqrt{11 - [x]^2}$.

Ans. $x \in [-3, -2) \cup [3, 4)$

Sol. $5 \leq [x]^2 \leq 11$

$\sqrt{5} \leq [x] \leq \sqrt{11} \qquad$ or $\qquad -\sqrt{11} \leq [x] \leq -\sqrt{5}$

$[x] = -3, 3$

$x \in [-3, -2) \cup [3, 4)$

3.1 RANGE BASED QUESTIONS

Find range

1. $f(x) = 16 \sin^2 x + 25 \operatorname{cosec}^2 x$.

Ans. $[41, \infty)$

Sol. $(4 \sin x)^2 + (5 \operatorname{cosec} x)^2 - 40 + 40$

$(4 \sin x - 5 \operatorname{cosec} x)^2 + 40$

$16 \sin^2 x = 25 \operatorname{cos ec}^2 x$

$\sin^4 x = \frac{25}{16}$

$y = [16 + 25, \infty)$

2. $f(x) = 4\sin^2 x + 6 \cos^2 x + 8 \sin x \cos x + 10$.

Ans. $[15 - \sqrt{17}, 15 + \sqrt{17}]$

Sol. $4 + 2 \cos^2 x + 4 \sin 2x + 10$

$15 + \cos 2x + 4 \sin 2x$

$15 + \left[-\sqrt{17}, \sqrt{17}\right]$

$x \in \left[15 - \sqrt{17}, 15 + \sqrt{17}\right]$

3. $f(x) = 9 \sec^2 x + 36 \operatorname{cosec}^2 x$.

Ans. $[81, \infty)$

Sol. $y = 9 + 9 \tan^2 x + 36 + 36 \cot^2 x$

$y = 45 + (3 \tan x)^2 + (6 \cot x)^2 + 36 - 36$

$y = 81 + (3 \tan x - 6 \cot x)^2$

$y \in [81, \infty)$

4. $f(x) = \frac{2x + 5}{3x - 4}, x \geq 0.$

Ans. $\left(-\infty, \frac{-5}{4}\right] \cup \left(\frac{2}{3}, \infty\right)$

Sol. Let $y = \frac{2x + 5}{3x - 4}$

$3xy - 4y = 2x + 5$

$x = \frac{4y + 5}{3y - 2} \geq 0$

$y \in \left(-\infty, -\frac{5}{4}\right] \cup \left(\frac{2}{3}, \infty\right)$

5. $f(x) = \frac{x^2 + 2x - 3}{x^2 - 4x + 3}$.

Ans. $R - \{1, -2\}$

Sol. $y = \frac{(x + 3)(x - 1)}{(x - 3)(x - 1)}$

$y = \frac{x+3}{x-3}, x \neq 1$

$y \in R - \{1, -2\}$

6. $f(x) = \frac{5}{3x+4}$

Ans. $R - \{0\}$

Sol. $y = \frac{5}{3x+4}$

$y \in R - \{0\}$

7. $f(x) = \frac{x^2+x+1}{x^2-x+1}$

Ans. $\left[\frac{1}{3}, 3\right]$

Sol. $y = \frac{x^2+x+1}{x^2-x+1}$

$(y-1)x^2 - (y+1)x + (y-2) = 0$

Case-I : $y \neq 1$

$D \geq 0$

$(y+1)^2 - 4(y-1)^2 \geq 0$

$(y+1+2y-2)(y+1-2y+2) \geq 0$

$(3y-1)(y-3) \leq 0$

$y \in \left[\frac{1}{3}, 3\right]$

Case-II : $y = 1$

$x = 0$

Which is real

$\Rightarrow y = 1$

8. $f(x) = 2\sec^2 x - 4$.

Ans. $[-2, \infty)$

Sol. $y = 2[1, \infty) - 4$

$y \in [2, \infty) - 4$

$y \in [-2, \infty)$

9. $f(x) = (2\sin x - 3)^2 - 1$

Ans. $[0, 24]$

Sol. $y \in [1, 25] - 1$

$y \in [0, 24]$

Case-(I) : f is monotonic

1. When x can be expressed in terms of y

$\frac{af(x)+b}{cf(x)+d}$

2. Range in case of implicit functions

3. Range in case of $f(g(x))$

Case 1 f is monotonic

$y = f(x), x \in [x_1, x_2]$

Q.1. $y = e^{2\sin x}$

Ans. $[e^{-2}, e^2]$

Sol. $y \in e^{[-2,2]}$

$y \in [e^{-2}, e^2]$

Q.2. $f(x) = \left(\frac{1}{2}\right)^{\cos^2 x}$

Ans. $\left[\frac{1}{2}, 1\right]$

Sol. $y = \left(\frac{1}{2}\right)^{[0,1]}$

$y \in \left[\frac{1}{2}, 1\right]$

Q.3. $f(x) = 2^{6x-x^2}$

Ans. $(0, 512]$

Sol. $y \in 2^{-\left(x^2-6x+9-9\right)}$

$y \in 2^{9-(x-3)^2}$

$y \in 2^{9+(-\infty,0]}$

$y \in (0, 512]$

Q.4. $f(x) = \ln(\sin x)$

Ans. $(-\infty, 0]$

Sol. $y \in \ell n\,(0, 1]$

$y \in (-\infty, 0]$

Q.5. $f(x) = \ln(x^2 - 2x + 2)$

Ans. $[0, \infty)$

Sol. $\ell n[1, \infty)$

$y \in [0, \infty)$

Case 2 f is non monotonic

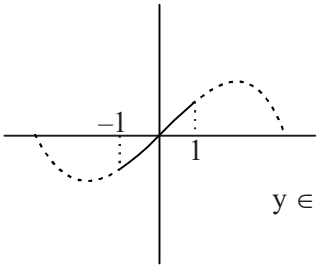
Q.1 $y = \sin x$

find range of y if

(i) $x \in [-1, 1]$

Ans. $[-\sin 1, \sin 1]$

Sol.

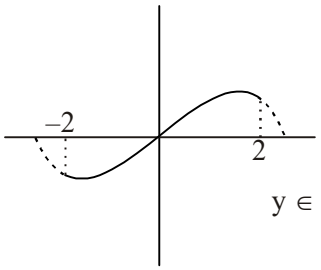


$y \in [-\sin 1, \sin 1]$

(ii) $x \in [-2, 2]$

Ans. $[-1,1]$

Sol.

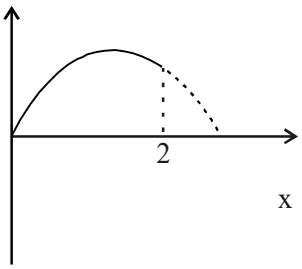


$y \in [-1,1]$

(iii) $x \in [0, 2]$

Ans. $[0,1]$

Sol.

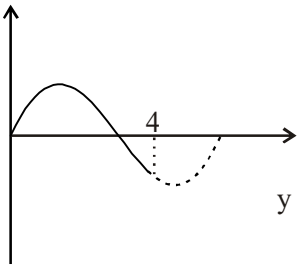


$x \in [0,1]$

(iv) $x \in [0, 4]$

Ans. $[\sin 4,1]$

Sol.

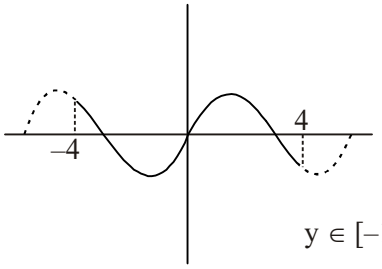


$y \in [\sin 4,1]$

(v) $x \in [-4, 4]$

Ans. $[-1,1]$

Sol



$y \in [-1,1]$

3.2 MISCELLANEOUS EXAMPLES ON RANGE

1. $f(x) = \cos^4 x - \sin^4 x$.

Ans. $f(x) \in [-1, 1]$

Sol. $f(x) = \cos 2x$

$y \in [-1, 1]$

2. $y = \cos^6 x + \sin^6 x$.

Ans. $f(x) \in \left[\frac{1}{4}, 1\right]$

Sol. $y = 1 - \frac{3}{4}(\sin 2x)^2$

$y \in 1 - \frac{3}{4}[0,1]$

$y \in 1 + \left[-\frac{3}{4}, 0\right]$

$y \in \left[\frac{1}{4}, 1\right]$

3. $y = x^8 + x^6 + 4x^2 + 7$.

Ans. $f(x) \in [7, \infty)$

4. $y = |\sin x| + |\cos x|$.

Ans. $f(x) \in \left[1, \sqrt{2}\right]$

Sol. $y = \sqrt{(|\sin x| + |\cos x|)^2}$

$y = \sqrt{1 + |\sin 2x|}$

$= \sqrt{1 + [0,1]}$

$= \sqrt{[1,2]}$

$= \left[1, \sqrt{2}\right]$

5. $y = \sqrt{x-2} + \sqrt{6-x}$.

Ans. $f(x) \in \left[2, 2\sqrt{2}\right]$

Sol. $y = \sqrt{\left(\sqrt{x-2} + \sqrt{6-x}\right)^2}$

$y = \sqrt{4 + 2\sqrt{(x-2)(6-x)}}$

$= \sqrt{4 + 2\sqrt{-x^2 + 8x - 12}}$

$= \sqrt{4 + 2\sqrt{-12 - \left(x^2 - 8x + 16 - 16\right)}}$

$= \sqrt{4 + 2\sqrt{4 - (x-4)^2}}$

$= \sqrt{4 + 2\sqrt{[0,4]}} = \sqrt{4 + [0,4]}$

$= \left[2, 2\sqrt{2}\right]$

6. $y = x(x+2)(x+4)(x+6)$.

Ans. $f(x) \in [-16, \infty)$

Sol. $y = (x^2 + 6x)(x^2 + 6x + 8)$

Let $x^2 + 6x = t$

$y = t(t + 8)$

$y = t^2 + 8t + 16 - 16$

$y = (t + 4)^2 - 16$

$y = (x^2 + 6x + 4)^2 - 16$

$y = (x^2 + 6x + 9 - 5)^2 - 16$

$y = [(x + 3)^2 - 5]^2 - 16$

$y = [0, \infty) - 16 = [-16, \infty)$

7. $y = \dfrac{8 + 3\sin x + 4\cos x}{8 - 3\sin x - 4\cos x}$.

Ans. $f(x) \in \left[\dfrac{3}{13}, \dfrac{13}{3}\right]$

Sol. $y_{max} = \dfrac{8+5}{8-5} = \dfrac{13}{3}$

$y_{min} = \dfrac{8-5}{8+5} = \dfrac{3}{13}$

8. $y = \dfrac{1}{\cos^4 x + \sin^4 x}$.

Ans. $f(x) \in [1, 2]$

Sol. $y = \dfrac{1}{1-\dfrac{1}{2}(\sin 2x)^2}$

$= \dfrac{1}{1-\dfrac{1}{2}[0,1]} = \dfrac{1}{\left[-\dfrac{1}{2},0\right]+1}$

$y \in [1, 2]$

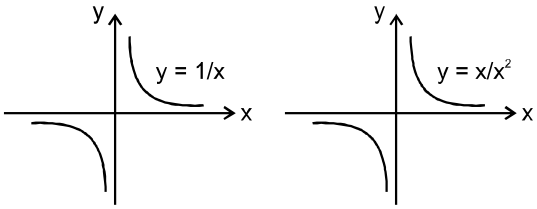
4. EQUAL/IDENTICAL FUNCTION

Two functions f and g are said to be identical (or equal) iff :

(i) The domain of f ≡ the domain of g.

(ii) f(x) = g(x), for every x belonging to their common domain.

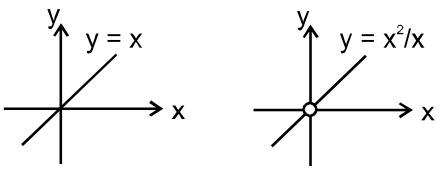
e.g. $f(x) = \dfrac{1}{x}$ and $g(x) = \dfrac{x}{x^2}$ are identical functions.



Clearly the graphs of f(x) and g(x) are exactly same

But f(x) = x and $g(x) = \dfrac{x^2}{x}$ are not identical functions.

Clearly the graphs of f(x) and g(x) are different at x = 0.



Determine whether following functions are identical

1. $f(x) = (\sqrt{x})^2$; g (x) = x.

Ans. No

Sol. $D_f = [0, \infty)$, $D_g = R$
No

2. $f(x) = \dfrac{1}{\sin x}$; g(x) = cosec x.

Ans. Yes

Sol. $D_f = D_g = R - \{n\pi\}$
f(x) = g(x) = cosec x
Yes

3. $f(x) = \dfrac{1}{\text{cosec}(x)}$, g (x) = sin x.

Ans. No

4. $f(x) = \dfrac{x^2}{x}$; g(x) = x.

Ans. No

Sol. $D_f = R - \{0\}$, $D_g = R$

No

5. $f(x) = \ln(e^x)$; $g(x) = e^{\ln x}$.

Ans. No

Sol. $D_f = R$ $\quad Dg \in (0, \infty)$

No

6. $f(x) = \sec^2 x - \tan^2 x$; $g(x) = \sin^2 x + \cos^2 x$.

Ans. No

Sol. $D_f \in R - (2n+1)\frac{\pi}{2}$, $D_g \in R$

NO

7. $f(x) = \log_x e$; $g(x) = \dfrac{1}{\log_e x}$.

Ans. Yes

Sol. $D_f = D_g = (0, 1) \cup (1, \infty)$

$f(x) = g(x) = \log_x e$

Yes

8. $f(x) = \sqrt{\dfrac{1-\cos 2x}{2}}$; $g(x) = \sin x$.

Ans. No

Sol. $f(x) = |\sin x|$, $g(x) = \sin x$

No

9. $f(x) = \sqrt{x+1}\ \sqrt{x-1}$; $g(x) = \sqrt{x^2-1}$

Ans. No

Sol. $D_f \in [1, \infty)$, $D_g \in (-\infty, -1] \cup [1, \infty)$

No

10. $f(x) = \sqrt{1+x}\ \sqrt{1-x}$; $g(x) = \sqrt{1-x^2}$.

Ans. Yes

Sol. $f(x) = g(x) = \sqrt{1-x^2}$

Yes

11. $f(x) = \dfrac{1}{1+\frac{1}{x}}$; $g(x) = \dfrac{x}{x+1}$.

Ans. No

Sol. $D_f = R - \{0, -1\}$, $D_g = R - \{-1\}$

No

12. $f(x) = x\,\text{sgn}(x)$; $g(x) = |x|$.

Ans. Yes

Sol. $D_f = D_g = R$

$f(x) = g(x) = |x|$

Yes

13. $f(x) = [\{x\}]$; $g(x) = \{[x]\}$.

Ans. Yes

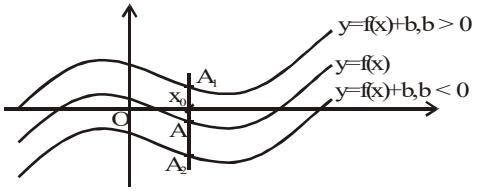
Sol. $D_f = D_g = R$

$f(x) = g(x) = 0$

Yes

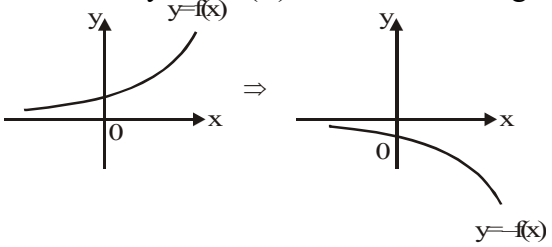
5. TRANSFORMATION OF GRAPHS

T - 4 Drawing the graph of y = f(x) + b, b ∈ R, from the known graph of y = f(x)



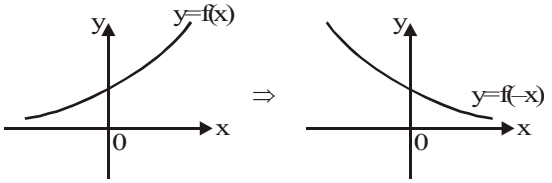
T - 5 Drawing the graph of y = –f(x) from the known graph of y = f(x)

To draw y = –f(x), take the image of the curve y = f(x) in the x-axis as plane mirror.



T - 6 Drawing the graph of y = f(–x) from the known graph of y = f(x)

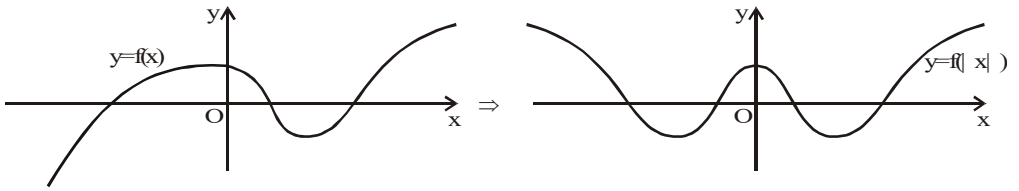
To draw y = f(–x), take the image of the curve y = f(x) in the y-axis as plane mirror.



T - 7 Drawing the graph of y = f(|x|) from the known graph of y = f(x)

It is clear that, $f(|x|) = \begin{cases} f(x), & x \geq 0 \\ f(-x), & x < 0 \end{cases}$.

Remove portion of the grapg lying on left of y-axis. Replace it with copy of mirror image of remaining portion of the graph wrt to y-axis

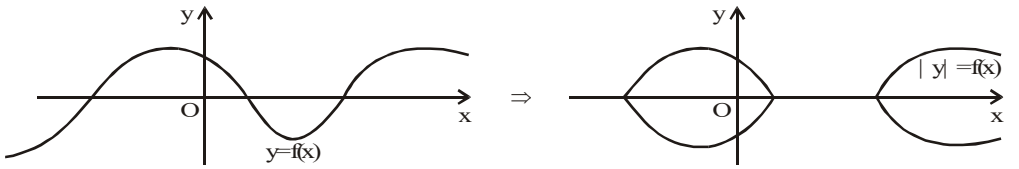


T -8 Drawing the graph of y = f(-|x|) from the known graph of y = f(x)

Remove portion of the graph lying on right of y-axis. Replace it with copy of mirror image of remaining portion of the graph wrt to y-axis

T -9 Drawing the graph of |y| = f(x) from the known graph of y = f(x)

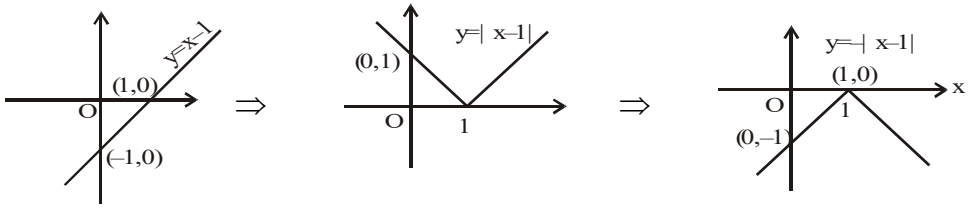
Remove portion of the graph lying below x-axis. Replace it with copy of mirror image of remaining portion of the graph wrt to x-axis



Ex.1 Draw the graph of $y = |2 - |x - 1||$.

Ans.

Sol.

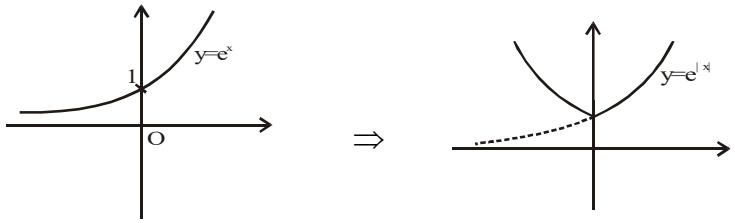


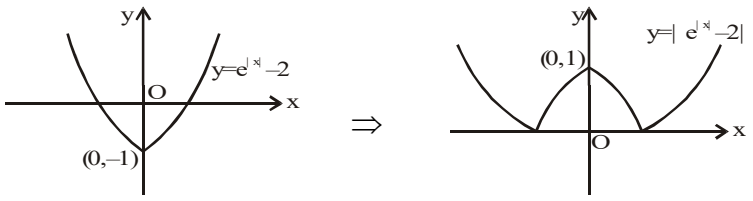


Ex.2 Draw the graph of $y = |e^{|x|} - 2|$

Ans.

Sol.

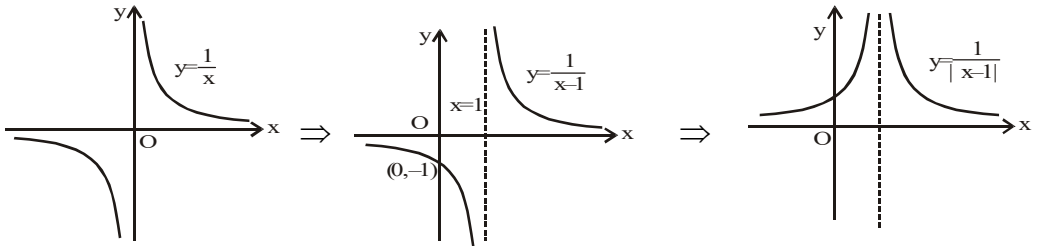


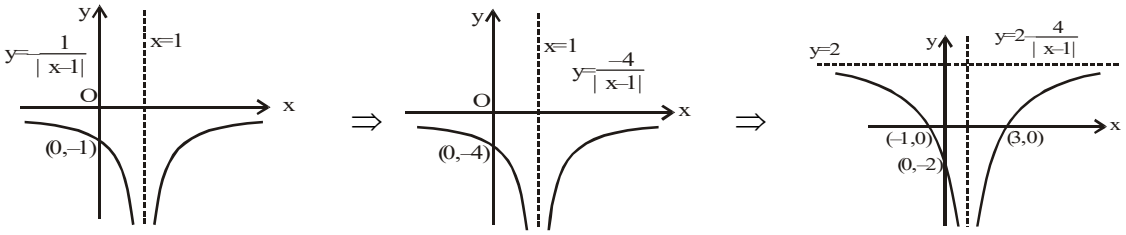


Ex.3 Draw the graph of $y = 2 - \frac{4}{|x-1|}$

Ans.

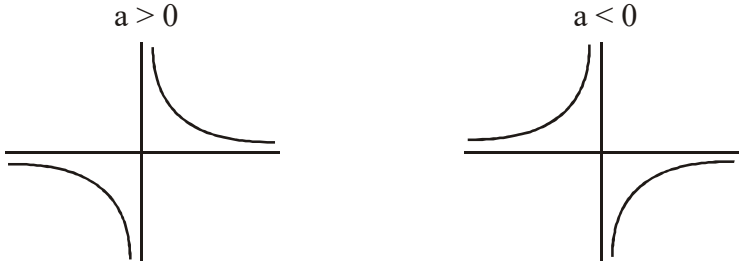
Sol.



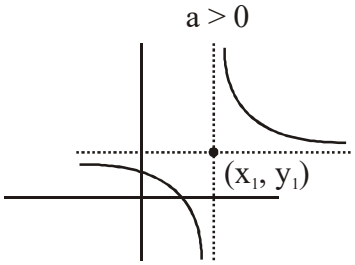


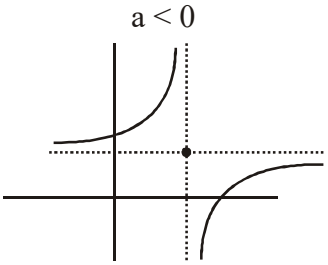
Note: Graph of $\frac{\text{linear}}{\text{linear}}$

$y = \frac{a}{x}$



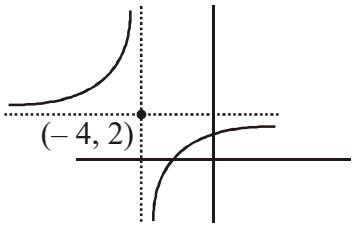
If function is $y - y_1 = \frac{a}{x - x_1}$ then shift the origin to (x_1, y_1)





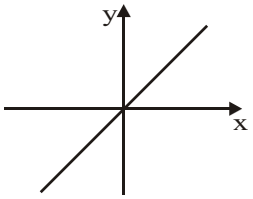
$$y = \frac{2x+1}{x+4} = \frac{2(x+4)-7}{x+4} = 2 - \frac{7}{x+4}$$

$$y - 2 = \frac{-7}{x+4} \Rightarrow \qquad (x_1, y_1) = (-4, 2)$$



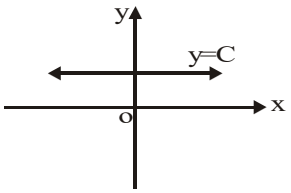
6. IMPORTANT DEFINITIONS

6.1 Identity function :



The function f : A $\rightarrow$ A defined by f(x) = x $\forall$ x $\in$ A is called the identity function on A and is denoted by I_A. It is easy to observe that identity function is a bijection.

6.2 Constant function :



f : A $\rightarrow$ B is said to be constant function if every element of A has the same f image in B. Thus

f : A $\rightarrow$ B ; f(x) = c, $\forall$ x $\in$ A, c $\in$ B is constant function.

Note that the range of a constant function is a singleton set.

Domain : R **Range :** {C}

6.3 Bounded function :

A function is said to be bounded if |f(x)| $\leq$ M, where M is a finite quantity.

In other words, if range of f(x) does not contain + ∞ or – ∞ then it will be bounded function.

6.4 Homogeneous functions

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables.

For examples $5x^2 + 3y^2 - xy$ is homogenous in x & y.

Symbolically if, $f(tx, ty) = t^n f(x, y)$ then f(x, y) is homogeneous function of degree n.

Q.1 $f(x, y) = \dfrac{x^2 - y^2}{x + 5y}$

Ans. Yes, Degree = 1

Q.2 $f(x, y) = \sin(xy)$

Ans. No

Q.3 $f(x, y) = x \ln\left(\dfrac{x}{y}\right) + y$

Ans. Yes, Degree = 1

6.5 Implicit & explicit function

If function has been expressed in form of y = f(x) then it is called explicit form otherwise it will be implicit form.

Convert following functions into explicit form.

1. $x^3 + y^3 = 1$

Ans. $y = \left(1 - x^3\right)^{1/3}$

2. $x = \dfrac{y+1}{y-3}$.

Ans. $y = \dfrac{3x+1}{x-1}$

3. $x = \dfrac{2^y}{2^y + 1}$.

Ans. $y = \log_2 \dfrac{x}{1-x}$

4. $x = \ln(y - 3)$.

Ans. $y = 3 + e^x$

5. $2^x + 2^y = 4$

Ans. $y = \log_2(4 - 2^x)$

7. CLASSIFICATION OF FUNCTION

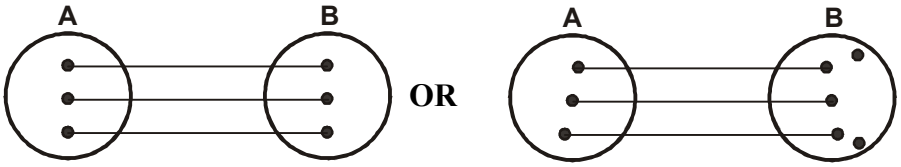
7.1 ONE-ONE AND MANY-ONE FUNCTION

One–One Function (Injective Mapping) :

If each element in the domain of a function has a distinct image in the co–domain the function is said to be one–one function and is also known as Injective Function.

$f: A \to B$ is one – one $\Leftrightarrow$ $a \neq b \Rightarrow f(a) \neq f(b)$ for all $a, b \in A$

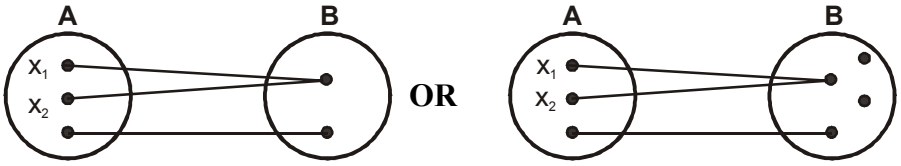
$f: A \to B$ is one – one $\Leftrightarrow$ $f(a) = f(b) \Rightarrow a = b$ for all $a, b \in A$



Many–One Function :

If there are two or more than two elements of domain having the same image then $f(x)$ is called Many–One function.

$f: A \to B$ is a many–one function, if there exist $x, y \in A$ such that $x \neq y$ but $f(x) = f(y)$.



Graphical Approach

A function is one to one if and only if a horizontal line intersects its graph at most once.

if a line parallel to x-axis cuts the graph of the function atleast at two points, then f is many-one.

Derivative Approach

If $\frac{dy}{dx}$ takes both positive and negative signs, then the function is many - one.

If function is one - one then, $\frac{dy}{dx} \geq 0$ or $\frac{dy}{dx} \leq 0$ (Always)

[provided equality doesn't always hold for an interval.]

Ex.1 $y = x + \sin x$ Ex.2 $y = 2x + \cos x$ Ex.3 $y = 3\sin x + x$

Standard Examples :

Examples of one - one function

1. Linear polynomial
2. $\left(\frac{\text{linear}}{\text{linear}}\right)$ with no common factor, $\frac{1}{\text{linear}}$
3. $y = a^x$, $y = \log_a x$

Examples of many - one function :

1. Even degree polynomial

2. $\frac{\text{quadratic}}{\text{quadratic}}, \frac{\text{linear}}{\text{quadratic}}, \frac{1}{\text{quadratic}}, \frac{\text{quadratic}}{\text{linear}}$ (with no common factor)

3. All trigonometric functions in their complete domain.

4. $y = \{x\}$, $y = [x]$, $y = |x|$, $y = \text{sgn}(x)$, etc. ([.] denotes G.I.F, {.} denotes fractional part function)

Determine nature of following functions (one-one or many - one)

1. $y = x - [x]$ ([.] denotes G.I.F)

Ans. many–one

Sol. $y = \{x\}$

2. $y = \frac{x+5}{x-3}$.

Ans. one–one

Sol. $\frac{\text{Linear}}{\text{Linear}}$

3. $y = 5 - 3x$.

Ans. one–one

4. $y = \frac{x^2-5x+6}{x^2-x-2}$.

Ans. one–one

Sol. $y = \frac{(x-2)(x-3)}{(x-2)(x+1)}$

5. $y = \frac{x^2+x+1}{x^2-x+1}$.

Ans. many–one

Sol. $y = \frac{\text{Quadratic}}{\text{Quadratic}}$

6. $y = 3\cos x - 2x$.

Ans. many–one

Sol. $\frac{dy}{dx} = -3\sin x - 2$

$= -3[-1,1] - 2$

$= [-3, 3] - 2$

$= [-5, 1]$

7. $y = 2\sin x + 5x$.

Ans. one–one

Sol. $\frac{dy}{dx} = 2\cos x + 5$

$= [-2,2] + 5$

$= [3, 7]$

8. $y = x^3 - 6x^2 + 12x - 7$.

Ans. one–one

Sol. $y = x^3 - 6x^2 + 12x - 7$

$$\frac{dy}{dx} = 3x^2 - 12x + 12$$

$$= 3(x^2 - 4x + 4) \geq 0$$

one–one

9. $y = x^4 + 3x^3 + 2x - 5$.

Ans. many–one

Sol. evern degree polynomial many–one.

10. $y = x^2 - 4x + 10$.

Ans. many–one

11. $y = |x| \, sgn(x) + sinx$.

Ans. one–one

Sol. $y = x + \sin x$

$y^1 = 1 + \cos x \geq 0, \ \forall \ x$

one–one

12. $y = \dfrac{1}{x^2 - 4x + 9}$.

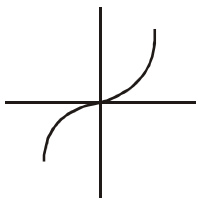
Ans. many–one

Sol. $\dfrac{1}{\text{Quadratic}} \Rightarrow$ Many one

13. $y = x|x|$.

Ans. one–one

Sol. $y = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$



one–one

14. If the function $f(x) = x^2 + bx + 3$ is not injective for values of x in the interval $0 \leq x \leq 1$ then b lies in

(A) $(-\infty, \infty)$ (B) $(-2, \infty)$ (C) $(-2, 0)$ (D) $(-\infty, 2)$

Ans. C

[Sol. For many one in [0, 1]

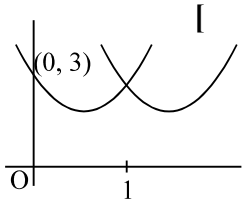
$0 <$ vertex < 1

$$0 < -\frac{b}{2} < 1$$

$0 < -b < 2$

$-2 < b < 0$

Note: If f is injective then $b \in R - (-2, 0)$]



15. $f: R \to R$ is defined as $f(x) = \begin{cases} x^2 + 2mx - 1 & \text{for } x \le 0 \\ mx - 1 & \text{for } x > 0 \end{cases}$.

If $f(x)$ is one-one then m must lies in the interval

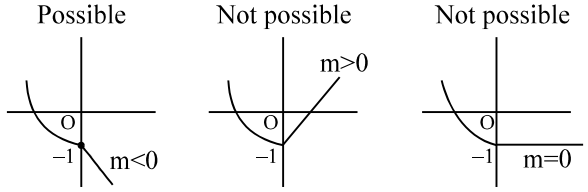
(A) $(-\infty, 0)$ (B) $(-\infty, 0]$ (C) $(0, \infty)$ (D) $[0, \infty)$

Ans. A

[Sol. for f to be one-one $f'(x) > 0$ or $f'(x) < 0$ for all x

clearly f is continuous at $x = 0$ and $f(0) = -1$

possible cases can be as shown



Note that for $m < 0$ the vertex of $f(x) = x^2 + 2mx - 1$ lie on + sides of x-axis]

16. Let a function f defined from $R \to R$ as: $f(x) = \begin{cases} x + m & \text{for } x \le 1 \\ 2mx - 1 & \text{for } x > 1 \end{cases}$

If the function is surjective on R then m must lie in the interval

(A) $(0, 2]$ (B) $(-\infty, 0]$ (C) $(-\infty, 0)$ (D) $(0, \infty)$

Ans. A

[Sol. for f to be surjective range must be set of all real numbers

as shown only figure-2 is possible

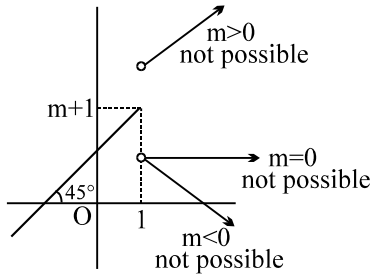


Figure-1

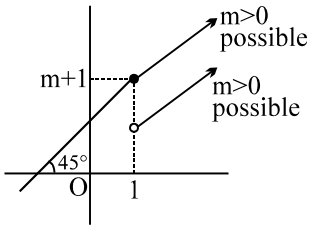


Figure-2

Hence $m + 1 \ge 2m - 1$

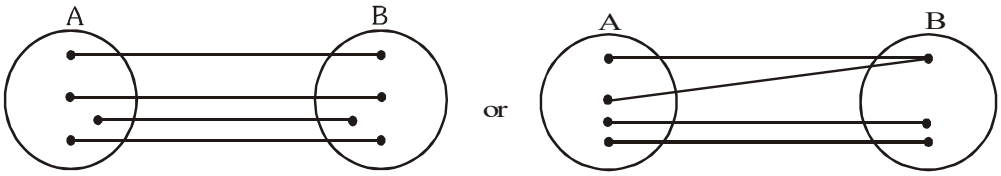
$2 \ge m \Rightarrow 0 < m \le 2$]

7.2 ONTO AND INTO FUNCTION

Onto function (Surjective mapping) :

If the function f : A $\to$ B is such that each element in B (co-domain) is the 'f' image of atleast one element in A, then we say that f is a function of A 'onto' B. Thus $f: A \to B$ is surjective if $\forall\ b \in B$, $\exists$ some $a \in A$ such that $f(a) = b$

Diagramatically surjective mapping can be shown

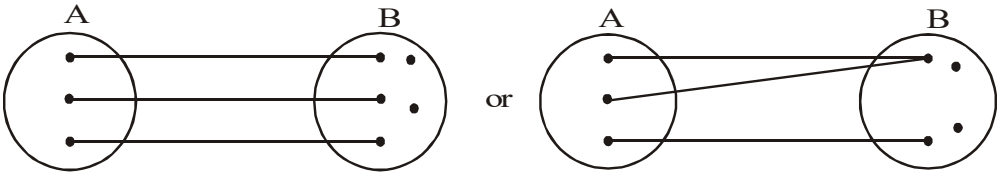


Note that : If range = co-domain, then f(x) is onto.

Into function :

If $f : A \to B$ is such that there exists atleast one element in co-domain which is not the image of any element in domain, then f(x) is into.

Diagramatically into function can be shown



Note :

1. For a function to be onto or into depends mainly on their co-domain.

Ex. $f : R \to R$ then $f(x) = |x|$ is a into function

$f : R \to R^+ \cup \{0\}$ then $f(x) = |x|$ is a onto function

2. $f : R \to R$ is a polynomial

(a) Of even degree, then it will neither be injective nor surjective.

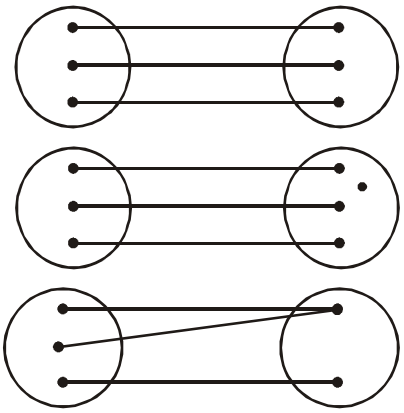
(b) Of odd degree, then it will always be surjective, no general comment can be given on its injectivity.

Thus a function can be one of these four types :

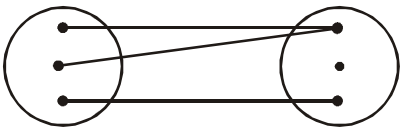
(i) one-one onto (injective & surjective)

(ii) one-one into (injective but not surjective)

(iii) many-one onto (surjective but not injective)



(iv) many-one into (neither surjective nor injective)



Note :

If 'f' is both injective & surjective, then it is called a **Bijective** mapping. The bijective functions are also named as invertible, non singular or biuniform functions.

Q.1 **Permutation based problems**

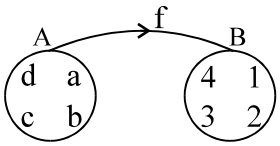
A $f: A \to B$

Case-I: When both the sets A and B contain an equal number of elements

(i) Total number of functions

Ans. $4^4 = 256$

(ii) Number of functions one-one



Ans. 4!

(iii) Number of functions many-one

Ans. $4^4 - 4!$

(iv) Number of onto function 4! $\Rightarrow$ number of bijective mapping

Ans. 4!

(v) Number of into functions

Ans. $4^4 - 4!$

Alternate : ${}^4C_1\,[3^4 - \{{}^3C_2 + {}^3C_1(2^4 - 2)\}] + {}^4C_2(2^4 - 2) + {}^4C_3 \cdot 1 = 4^4 - 4!\ ]$

Case-II: When number of elements in A(domain) is more than B

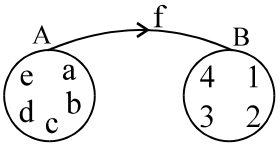
(i) Total number of functions

Ans. $4^5 = 1024$

(ii) One-one (injective)

Ans. Nil

(iii) Many-one



Ans. 1024

(iv) Number of onto function

Ans. $\left(\frac{5!}{1!\cdot 1!\cdot 2!\cdot 3!}\right) 4! = 240$ (5 books in 4 groups 1, 1, 1, 2)

(v) Number of into function

Ans. $1024 - 240 = 784$

Case-III: Number of elements in codomain (B) is more than A

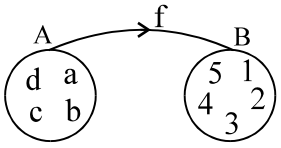
(i) Total functions

Ans. $5^4 = 625$

(ii) Number of injective mapping

Ans. ${}^5C_4 \cdot 4! = 120$

(iii) Number of many-one



Ans. $625 - 120 = 505$

(iv) Number of onto function

Ans. 0

(v) Number of into function

Ans. 625

8. FUNCTIONAL EQUATIONS

8.1 Type -1

1. $f(2x + 1) = x + 4$ find $f(x)$.

Ans. $f(x) = \frac{x+7}{2}$

Sol. Let $2x + 1 = t$

$$x = \frac{t-1}{2}$$

$$f(t) = \frac{t-1}{2} + 4 = \frac{t+7}{2}$$

2. $f\left(\frac{x+1}{x-2}\right) = x + 4$ find $f(x)$.

Ans. $f(x) = \frac{3(2x-1)}{x-1}$

Sol. Let $\frac{x+1}{x-2} = t$

$x + 1 = xt - 2t$

$$x = \frac{1+2t}{t-1}$$

$$f(t) = \frac{2t+1}{t-1} + 4 = \frac{3(2t-1)}{t-1}$$

8.2 Type - 2

1. $2f(x) + f(1-x) = 2x - 1, \forall x \in R$ find $f(x)$.

Ans. $f(x) = 2x - 1$

Sol. $2f(x) + f(1 - x) = 2x - 1$ ...(1)

$2f(x - 1) + f(x) = 1 - 2x$ ...(2)

$2(1) - (2)$

$3f(x) = 4x - 2 - 1 + 2x$

$= 6x - 3$

$f(x) = 2x - 1$

2. $3f(x) - 2f\left(\frac{1}{x}\right) = x + 1 \ \forall x \in R - \{0\}$ find $f(x)$.

Ans. $f(x)=\frac{3x^2+5x+2}{5x}$

Sol. $3f(x)-2f\left(\frac{1}{x}\right)=x+1$ ...(1)

$x \to \frac{1}{x}$

$3f\left(\frac{1}{x}\right)-2f(x)=\frac{1}{x}+1$ ...(2)

$3\times(1)+2\times(2)$

$5f(x)=3x+3+\frac{2}{x}+2$

$=\frac{3x^2+5x+2}{5x}$

8.3 Type - 3

1. $f(x)=ax^5+bx^3+cx+d\sin x+4$, $f(3)=7$; find $f(-3)$

Ans. 1

Sol. $f(x)+f(-x)=8$

$x=3$

$7+f(-3)=8$

$f(-3)=1$

2. $f(x)=ax^4+bx^2+c|x|+7x$, $f(-2)=5$ find $f(2)$.

Ans. 33

Sol. $f(x)-f(-x)=14x$

$x=2$

$f(2)-5=14\times 2$

$f(2)=33$

8.3 Type - 4

1. $f(x+y)=f(x)+g(y)+8\ \forall\ x, y\in R$ $f(x)=x\ \forall\ x<0$; $g(8)=15$ find $f(0)$.

Ans. $f(0)=16$

Sol. Put $x=8$

$f(8+y)=15+f(y)+8$

$y=-8$

$f(0)=15-8$

2. $f(x)=x^5+ax^4+bx^3+cx^2+dx+e$; $f(i)=i\ \forall\ i=1,2,3,4,5$ find $f(0)$.

Ans. $f(0)=-120$

Sol. Let $h(x)=f(x)-x$

$\Rightarrow f(x)-x=(x-1)(x-2)(x-3)(x-4)(x-5)$

put $x=0$

$f(0)=-120$

9.1 COMPOSITE OF TWO FUNCTIONS

Let $f : A \to B$ & $g : B \to C$ be two functions. Then the function $gof : A \to C$ defined by $(gof)(x) = g(f(x)) \ \forall \ x \in A$ is called the composite of the two functions f & g.

Diagramatically $\xrightarrow{x}$ [f] $\xrightarrow{f(x)}$ [g] $\longrightarrow g(f(x))$

Example $f(x) = \sin x$ $g(x) = \ln x$

1. $gof(x) = g(f(x)) = \ln(f(x)) = \ln(\sin x)$
2. $fog(x) = f(g(x)) = \sin(g(x)) = \sin(\ln x)$
3. $fof(x) = f(f(x)) = \sin(f(x)) = \sin(\sin x)$
4. $gog(x) = g(g(x)) = \ln(g(x)) = \ln(\ln x)$

Q.1. $f(x) = \sqrt{x-4}$ $g(x) = x^2$ find domain of
(i) fog (ii) gof (iii) fof (iv) gog

Ans. (i) $x \in (-\infty, -2] \cup [2, \infty)$ (ii) $x \in [4, \infty)$ (iii) $x \in [20, \infty)$ (iv) $x \in R$

Sol. (i) $f\left(g(x)\right) = \sqrt{x^2 - 4}$

(ii) $g\left(f(x)\right) = \left(\sqrt{x^2 - 4}\right)^2$

(iii) $f\left(f(x)\right) = \sqrt{\sqrt{x-4} - 4}$

$\sqrt{x-4} \geq 4$ & $x \geq 4$

$x \geq 20$

(iv) $g(g(x)) = x^4$

Q.2. $f(x) = \sqrt{x}$ $g(x) = \sqrt{5-x}$ find domain of.
(i) fog (ii) gof (iii) fof (iv) gog

Ans. (i) $x \leq 5$ (ii) $x \in [0, 25]$ (iii) $x \in [0, \infty)$ (iv) $x \in [-20, 5]$

Sol. (i) $f\left(g(x)\right) = \sqrt{\sqrt{5-x}}$

(ii) $g\left(f(x)\right) = \sqrt{5-\sqrt{x}}$

$5 - \sqrt{x} \geq 0$ & $x \geq 0$

$x \leq 25$

(iii) $f\left(f(x)\right) = \sqrt{\sqrt{x}}$

(iv) $g\left(g(x)\right) = \sqrt{5-\sqrt{5-x}}$

$5 \geq \sqrt{5-x}$ & $5 \geq x$

$5 - x \leq 25$

$x \geq -20$

Q.3. $f(x) = 2x + 1$ $g(f(x)) = 4x^2 + 6x + 7$ find $g(x)$.

Ans. $g(x) = x^2 + x + 5$

Sol. $g(f(x)) = 4x^2 + 4x + 1 + 2x + 1 + 5$

$g(f(x)) = (2x + 1)^2 + (2x + 1) + 5$

$g(x) = x^2 + x + 5$

Q.4 $f(x)=\dfrac{2x-1}{x+5}$; $gof(x)=x$; find $g(x)$

Ans. $g(x)=\dfrac{5x+1}{2-x}$

Sol. $xf(x)+5f(x)=2x-1$

$$x=\frac{5f(x)+1}{2-f(x)}=g(f(x))$$

Q.5 $f(x)=x^x$; $g(x)=x^{2x}$ then $fog(x)$ will be.

(A) $x^{x^{2x+1}}$ (B) $x^{2x^{2x}}$ (C) $x^{2x^{2x+1}}$ (D) $x^{2x^{4x}}$

Ans. C

Sol. $f(g(x))=(g(x))^{(g(x))}$

$$=\left(x^{2x}\right)^{x^{2x}}=x^{2x^{2x+1}}$$

Q.6 Let $f(x)=\dfrac{x-1}{x+1}, x\in R-\{0,-1,1\}$. If $f^{n+1}(x)=f\left(f^n(x)\right)$ for all $n\in N$, then $f^6(6)+f^7(7)$ is equal to:

माना $f(x)=\dfrac{x-1}{x+1}, x\in R-\{0,-1,1\}$ है। यदि $f^{n+1}(x)=f\left(f^n(x)\right)$, सभी $n\in N$ के लिए है, तो $f^6(6)+f^7(7)$ बराबर है :

Ans. $-\dfrac{3}{2}$

Sol. $f(x)=\dfrac{x-1}{x+1}$

$$\Rightarrow f^2(x)=f(f(x))=\frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1}=-\frac{1}{x}$$

$$f^3(x)=f\left(f^2(x)\right)=f\left(-\frac{1}{x}\right)=\frac{x+1}{1-x}$$

$$\Rightarrow f^4(x)=f\left(\frac{x+1}{1-x}\right)=-\frac{1}{x}$$

$$\Rightarrow f^6(x)=-\frac{1}{x}\Rightarrow f^6(6)=-\frac{1}{6}$$

$$f^7(x)=f\left(-\frac{1}{x}\right)=\frac{x+1}{1-x}$$

$$\Rightarrow f^7(7)=\frac{8}{-6}=-\frac{4}{3}$$

$$\therefore -\frac{1}{6}+-\frac{4}{3}=-\frac{3}{2}$$

Q.7 Let f(x) and g(x) be two polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of f(2) + g(2) is _____.

माना दो वास्तविक बहुपद f(x) तथा g(x) क्रमशः घात 2 तथा 1 के हैं। यदि $f(g(x)) = 8x^2 - 2x$ तथा $g(f(x)) = 4x^2 + 6x + 1$ हैं, तो f(2) + g(2) का मान है _____.

Ans. 18

Sol. $f(x) = ax^2 + bx + c$

$g(x) = dx + e$

$fog(x) = a(g(x))^2 + bg(x) + c$

$= a[d^2x^2 + e^2 + 2dex] + b[dx + e] + c$

$= ad^2x^2 + x(2ade + bd) + e^2 + be + c$

$gof(x)$

$= df(x) + e$

$= adx^2 + bdx + cd + e$

on comparing

$ad = 4, bd = 6, cd + e = 1$

$ad^2 = 8, 2ade + bd = -2$

$e^2 + be + c = 0$

$ae = -2$

On solving

$a = 2, b = 3, c = 1, d = 2, e = -1$

$f(x) = 2x^2 + 3x + 1$

$g(x) = 2x - 1$

Properties

1. The composite of functions is not commutative i.e. $gof \neq fog$.
2. gof is defined only if $\forall\ x \in A$, f(x) is an element of the domain of 'g' so that we can take its g-image. Hence in gof(x) the range of 'f' must be a subset of the domain of 'g'.
3. The composite of functions is associative i.e. if f, g, h are three functions such that fo(goh) & (fog)oh are defined, then fo(goh) = (fog)oh.

   Associativety: $f : (N) \to I_0 \quad f(x) = 2x$

   $g : I_0 \to Q \quad g(x) = \frac{1}{x}$

   $h : Q \to R \quad h(x) = e^{\frac{1}{x}}$

   $(hog)of = ho(gof) = e^{2x}$
4. The composite of two bijections is a bijection i.e. if f and g are two bijections such that gof is defined, then gof is also a bijection.
5. If function f(x) is many – one then gof(x) will be many – one. If gof(x) is one-one then f(x)

will be one - one.

6. If g(x) is into then gof(x) will be into. If gof(x) is onto then g(x) will be onto.

9.2 COMPOSITE OF NON-UNIFORM FUNCTIONS

Let $f(x) = \begin{cases} x+1, & x \le 1 \\ 2x+1, & 1 < x \le 2 \end{cases}$ and $g(x) = \begin{cases} x^2, & -1 \le x < 2 \\ x+2, & 2 \le x \le 3 \end{cases}$, find (fog)

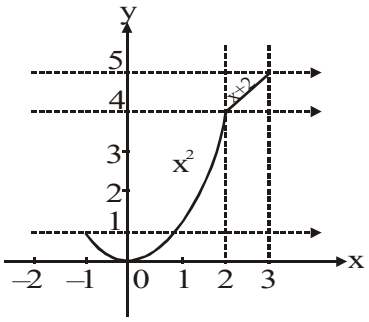
Solution : $f(g(x)) = \begin{cases} g(x)+1, & g(x) \le 1 \\ 2g(x)+1, & 1 < g(x) \le 2 \end{cases}$

Here, g(x) becomes the variable that means we should draw the graph.

It is clear that $g(x) \le 1$; $\forall\ x \in [-1, 1]$

and $1 < g(x) \le 2$; $\forall\ x \in (1, \sqrt{2}\,]$

$\Rightarrow$ $f(g(x)) = \begin{cases} x^2+1, & -1 \le x \le 1 \\ 2x^2+1, & 1 < x \le \sqrt{2} \end{cases}$



10.1 ODD AND EVEN FUNCTION

A function f(x) defined on the symmetric interval (–a, a)

1. If $f(-x) = f(x)$ for all x in the domain of 'f' then f is said to be an even function. e.g. $f(x) = \cos x$; $g(x) = x^2 + 3$.
2. If $f(-x) = -f(x)$ for all x in the domain of 'f' then f is said to be an odd function. e.g. $f(x) = \sin x$; $g(x) = x^3 + x$.

Properties

1. A function may neither be odd nor even .
2. Every even function is symmetric about the y–axis & every odd function is symmetric about the origin .
3. Derivative of an even function is an odd function.
4. Derivative of an odd function is an even function.
5. Every function defined on symmetric interval can be expressed as the sum of an even & an odd function.

$$f(x) = \underbrace{\boxed{\frac{f(x) + f(-x)}{2}}}_{\text{EVEN}} + \underbrace{\boxed{\frac{f(x) - f(-x)}{2}}}_{\text{ODD}}$$

6. The only function which is defined on the entire number line and is even and odd at the same time is $f(x) = 0$.
7. Even functions are generally many-one.

Determine nature of following functions(Odd, Even or NENO)

1. $f(x) = \sqrt{1+x} + \sqrt{1-x}$.

Ans. Even

Sol. $f(-x)=\sqrt{1-x}+\sqrt{1+x}=f(x)$

2. $f(x)=x^3\sin x$

Ans. Even

Sol. $f(-x)=(-x)^3\cdot\sin(-x)=f(x)$

3. $f(x)=x^2+x\sin x.$

Ans. Even

Sol. $f(-x)=(-x)^2+(-x)\cdot\sin(-x)=f(x)$

4. $f(x)=x^4+2x.$

Ans. Neno

Sol. $f(-x)=(-x)^4+2(-x)=x^4-2x$

5. $f(x)=x^3-5x.$

Ans. Odd

Sol. $f(-x)=(-x)^3-5(-x)=-f(x)$

6. $f(x)=\sin x+\cos x.$

Ans. Neno

Sol. $f(-x)=\sin(-x)+\cos(-x)=-\sin x+\cos x$

7. $f(x)=\sqrt{1+x}-\sqrt{1-x}\,.$

Ans. Odd

Sol. $f(-x)=\sqrt{1-x}-\sqrt{1+x}=-f(x)$

8. $f(x)=\ln\left(\frac{1+x}{1-x}\right).$

Ans. Odd

Sol. $f(-x)=\ell n\left(\frac{1-x}{1+x}\right)=-f(x)$

9. $f(x)=\frac{2^x+1}{2^x-1}.$

Ans. Odd

10. $f(x)=\ln(x+\sqrt{1+x^2})$

Ans. Odd

Sol. $f(-x)=\ell n\left(\sqrt{1+x^2}-x\right)=-f(x)$

11. $f(x)=x^2-|x|.$

Ans. Even

Sol. $f(-x)=(-x)^2-|-x|=f(x)$

12. $f(x)=\frac{(2^x+1)^2}{2^x}.$

Ans. Even

Sol. $$f(-x)=\frac{\left(1+2^{x}\right)^{2}}{\left(2^{x}\right)^{2}\times\frac{1}{\left(2^{x}\right)}}=f(x)$$

13. $f(x)=\cos(\sin x)$

Ans. Even

Sol. $f(-x)=\cos(\sin(-x))$
$=\cos(-\sin x)=f(x)$

14. $f(x)=\sin(\cos x)$

Ans. Even

Sol. $f(-x)=\sin(\cos(-x))=f(x)$

15. $f(x)=\sin(\tan x)$

Ans. Odd

Sol. $f(-x)=\sin(\tan(-x))=-\sin(\tan x)=-f(x)$

16. $f(x)=x+\{x\}$

Ans. Neno

Sol. $f(-x)=-x+\{-x\}$ = NENO

10.2 INVERSE OF A FUNCTION

For a function to be invertible, it must be one - one and onto.

If $f: X \to Y$ be a function defined by $y = f(x)$ such that ' f ' is both one–one and onto (bijective function), then there exists a unique function $g: Y \to X$ such that for each $y \in Y$, $g(y) = x$. The function $g(x)$ so defined is called the inverse of ' f ' and

Inverse of f is denoted by f^{-1}.

How to find inverse of f

Ex.1 Find inverse of $f(x) = \ln x$

Sol Replace $f(x)$ by y $\quad y = \ln x$

$x \leftrightarrow y$

$x = \ln y$

This is implicit form of inverse.

So, convert it into explicit form.

$$f^{-1}(x) = y = e^{x}$$

Properties

1. The inverse of a bijection is unique.

2. The inverse of a bijection is also a bijection.

3. Inverse of the inverse of the function is the function itself

4. Domain of f = Range of f^{-1}; Range of f = Domain of f^{-1}

5. If $f: A \to B$ is a bijection & $g: B \to A$ is the inverse of f, then

(i) $fog = I_B$ and $gof = I_A$, where I_A & I_B are identity functions on the sets A & B respectively.

(ii) The graphs of f & g are the mirror images of each other in the line $y = x$.

As shown in the figure given below a point (x ', y') corresponding to $y = x^2 (x \geq 0)$ changes to (y', x') corresponding

to $y = +\sqrt{x}$, the changed form of $x = \sqrt{y}$.

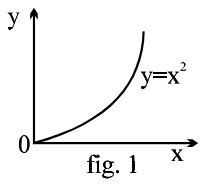


fig. 1

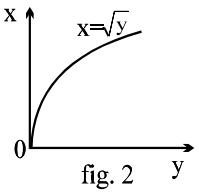


fig. 2

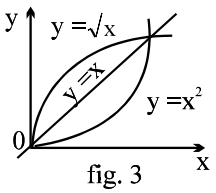


fig. 3

6. If f & g are two bijections $f: A \to B$, $g: B \to C$ then the inverse of gof exists and $(gof)^{-1} = f^{-1}og^{-1}$.
7. If f(x) and g(x) are continuous functions and inverse of each other then they will intersect at $y = x$ if they intersect.

Q.1. Find inverse of following functions.

(i) $f(x) = \dfrac{x+5}{3x-4}$ (ii) $f(x) = e^x + 2$ (iii) $f(x) = 2x + 5$ (iv) $f(x) = \ln(x + \sqrt{x^2+1})$

Ans. (i) $f^{-1}(x) = \dfrac{4x+5}{3x-1}$ (ii) $f^{-1}(x) = \ln(x-2)$

(iii) $f^{-1}(x) = \dfrac{x-5}{2}$ (iv) $f^{-1}(x) = \dfrac{e^{2x}-1}{2e^x}$

Sol. (i) $x \leftrightarrow y$

$$x = \frac{y+5}{3y-4}$$

$$y = \frac{4x+5}{3x-1}$$

(ii) $x \leftrightarrow y$

$x = e^y + 2$

$y = \ell n(x-2)$

(iii) $x = 2y + 5$

$$y = \frac{x-5}{2}$$

(iv) $x = \ell n\left(y + \sqrt{y^2+1}\right)$

$$\sqrt{y^2+1} + y = e^x$$

$$\sqrt{y^2+1} - y = e^{-x}$$

$$2y = \frac{e^x - e^{-x}}{2}$$

$$y = \frac{e^{2x}-1}{2e^x}$$

Q.2 $f: [2, \infty) \to [1, \infty)$ $f(x) = x^2 - 4x + 5$ find $f^{-1}(x)$.

Ans. $f^{-1}(x) = 2 + \sqrt{x-1}$

Sol. $x = y^2 - 4y + 5$

$(y-2)^2 = x - 1$

$y = 2 + \sqrt{x-1},\ 2 - \sqrt{x-1}$

Rejected

Q.3 $f(x) = x^3 + 3x + 2$ find $f^{-1}(6)$

Ans. 1

Sol. Let $f^{-1}(6) = k$

$\Rightarrow f(k) = 6$

$\Rightarrow k = 1$

Q.4 Let f: R–{3} $\to$ R – {1} be defined by $f(x) = \frac{x-2}{x-3}$ Let g : R $\to$ R be given as g(x) = 2x –3. Then the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to. **[JEE(Main)Mar.2021]**

माना f: R–{3} $\to$ R – {1}, $f(x) = \frac{x-2}{x-3}$ द्वारा परिभाषित है। माना g : R $\to$ R, g(x) = 2x –3 द्वारा दिया गया है। तो x के सभी मानों, जिनके लिए $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ है, का योगफल बराबर है : **[JEE(Main)Mar.2021]**

Ans. 5

Sol. $y = \frac{x-2}{x-3}$ $\qquad$ $y = 2x - 3$

$xy - 3y = x - 2$ $\qquad$ $x = \frac{y+3}{2}$

$x(y-1) = 3y - 2$ $\qquad$ $g^{-1}(x) = \frac{x+3}{2}$

$x = \frac{3y-2}{y-1}$

$f^{-1}(x) = \frac{3x-2}{x-1}$

$f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$

$\frac{3x-2}{x-1} + \frac{x+3}{2} = \frac{13}{2}$

$\frac{6x - 4 + (x+3)(x-1)}{(x-1)} = 13$

$6x - 4 + x^2 + 2x - 3 = 13x - 13$

$x^2 + 8x - 7 - 13x + 13 = 0$

$x^2 - 5x + 6 = 0$

$(x - 2)(x - 3) = 0$

$x = 2, 3$

sum $= 2 + 3 = 5$

Q.5 For some a, b, c $\in$ N, let $f(x) = ax - 3$ and $g(x) = x^b + c$, $x \in R$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$, then (fog) (ac) + (gof) (b) is equal to _____.

किसी a, b, c $\in$ N के लिए, माना $f(x) = ax - 3$ तथा $g(x) = x^b + c$, $x \in R$ है। यदि $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$ है, तो (fog) (ac) + (gof) (b) बराबर है।

Ans. 2039

Sol. Let fog(x) = h(x)

$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$

$\Rightarrow h(x) = fog(x) = 2x^3 + 7$

$fog(x) = a(xb + c) - 3$

$\Rightarrow a = 2, b = 3, c = 5$

$\Rightarrow fog(ac) = fog(10) = 2007$

$g(f(x) = (2x - 3)^3 + 5$

$\Rightarrow gof(b) = gof(3) = 32$

$\Rightarrow$ sum = 2039

10.3 PERIODIC FUNCTIONS

A function $f(x)$ is said to be periodic function if, there exists a fixed positive real number T independent of x, such that, $f(x + T) = f(x)$ $\forall$ $x \in$ Domain.

T is called one of the period of the function

Here the least positive value of T is called the fundamental period of the function.

Clearly $f(x) = f(x + T) = f(x + 2T) = f(x + 3T) \ldots$

If domain of function is bounded then function will be aperiodic.

Points where function is not defined should also follow periodic behaviour.

Important Periods

- sin x, cos x, sec x and cosec x are periodic functions with period 2π.
- tan x and cot x are periodic functions with period π.
- |sinx|, |cosx|, |tanx|, |cotx|, |secx|, |cosecx| are periodic functions with period π.

- $\sin^n x$, $\cos^n x$, $\sec^n x$, $\operatorname{cosec}^n x$ are periodic functions with period 2π or π according as n is odd or even.
- $\tan^n x$ and $\cot^n x$ are periodic function with period π whether n is odd or even.
- Period of $\{x\}$ is 1

Properties

1. Periodic functions are many - one.
2. A constant function is periodic but does not have a fundamental period.
3. If f(x) is periodic with period T, then

- a.f(x) is periodic with period T.
- f(x + c) is periodic with period T.
- f(x) ± c is periodic with period T.
- b f(ax + b) has period $\frac{T}{|a|}$, i.e., period is affected only by coefficient of 'x' where; a, b, c are constants with $a, b \neq 0$.

4. If f(x), g(x) are periodic functions with periods T_1, T_2 respectively then; we have

$h(x) = f(x) \pm g(x)$; $h(x) = f(x)g(x)$ or $h(x) = \frac{f(x)}{g(x)}$ has period as,

- LCM of $\{T_1, T_2\}$;

We will check at half of LCM also.

While taking LCM we should always remember,

- LCM of p and q always exist if p/q is a rational quantity.
- LCM of rational with rational is possible

Example :

$$\text{LCM of } \left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM. of } (a, c, e)}{\text{HCF of } (b, d, f)}$$

Where $a, b, c, d, e, f \in I$ with $b, d, f \neq 0$

- LCM of irrational with irrational is possible (but not always)
- But LCM of rational and irrational is not possible.

e.g., LCM of $(2\pi, 1, 6\pi)$ is not possible as $2\pi, 6\pi$ are irrational and '1' is rational.

5. If 'g' is periodic then

(i) fog will always be a periodic function. Period of fog may or may not be the period of 'g'.

(ii) gof may or may not be periodic

Find fundamental period of following functions

1. $f(x) = \sin^4\left(\frac{x-\pi}{7}\right)$.

Ans. 7π

Sol. $T = 7 \times \pi$

2. $f(x) = \cos(\cos x)$

Ans. π

Sol. $\cos(\cos(\pi + x)) = \cos(-\cos x) = \cos(\cos x)$

3. $f(x) = \sin(\cos x)$

Ans. 2π

Sol. $\sin(\cos x)$

$\downarrow$

2π

$\sin(\cos(\pi + x)) = -\sin(\cos x) \neq f(x)$

$T = 2\pi$

4. $f(x) = \sin x + \{x\}$

Ans. operiodic

Sol. $f(x) = \sin x + \{x\}$

$\downarrow \quad \downarrow$

LCM $(2\pi, \quad 1)$ = Not defined

5. $f(x) = |\sin x| + |\cos x|$

Ans. $\pi / 2$

Sol. $f(x) = |\sin x| + |\cos x|$

$\downarrow \quad \downarrow$

LCM $\{\pi,\pi\} = \pi$

$$f\left(\frac{\pi}{2} + x\right) = f(x)$$

$$T = \frac{\pi}{2}$$

6. $f(x) = \sin^2\left(\frac{2x}{3}\right) + \cos^2\left(\frac{3x}{5}\right)$.

Ans. 15π

Sol. $\text{LCM}\left\{\frac{3\pi}{2}, \frac{5\pi}{3}\right\} = 15\pi$

7. $f(x) = [x] + [2x] + [3x] - 6x + 7$.

Ans. 1

Sol. $f(x) = x - \{x\} + 2x - \{2x\} + 3x - \{3x\} - 6x + 3$

$= 7 - \{x\} - \{2x\} - \{3x\}$

$$\text{LCM}\left\{1, \frac{1}{2}, \frac{1}{3}\right\} = 1$$

8. $f(x) = \cos(\sin x) + \cos(\cos x)$

Ans. $\frac{\pi}{2}$

Sol. $\cos(\sin x) + \cos(\cos x)$

LCM $\{\pi, \pi\} = \pi$

Check $f\left(\frac{\pi}{2} + x\right) = f(x)$

$T = \frac{\pi}{2}$

9. $f(x) = \sin\sqrt{x}$.

Ans. operiodic

Sol. $\sin\sqrt{2\pi + x} \neq \sin\sqrt{x}$

10. $f(x) = \sin^2\left(\frac{2x}{3}\right), x \neq n\pi$

Ans. 3π

Sol. $\sin^2\left(\frac{2x}{3}\right), x \neq n\pi$

$\text{LCM}\left\{\frac{3\pi}{2}, \pi\right\} = 3\pi$

11. $f(x-3) = f(x+6)$

Ans. 9

Sol. $f(x-3) = f(x+6)$

$x \to x + 3$

$f(x) = f(x+9)$

$T = 9$

12. $f(x) + f(x+4) + f(x+8) = 0$

Ans. 12

Sol. $f(x) + f(x+4) + f(x+8) = 0$ ...(1)

$x \to x + 4$

$f(x+4) + f(x+8) + f(x+12) = 0$ ...(2)

(1) – (2)

$f(x) = f(x+12)$

$T = 12$

13. Let f is periodic with period $\frac{1}{2}$. If $f(2) = 5$ and $f\left(\frac{9}{4}\right) = 2$. Find $f(-3)$ and $f\left(\frac{1}{4}\right)$.

Ans.

[Sol. $f\left(\frac{9}{4}\right) = 2 \quad \Rightarrow \quad f\left(2 + \frac{1}{4}\right) = 2 \quad \Rightarrow \quad f\left(\frac{1}{4}\right) = 2$

14. Consider those functions f that satisfy $f(x+4) + f(x-4) = f(x)$ for all real x. Any such function is periodic, and there is a least common positive period p for all of them. The value of p, is

(A) 8 (B) 12 (C) 16 (D) 24

Ans. D

Sol. $f(x+4) + f(x-4) = f(x)$ _____(i)

Replacing x by $x + 4$, we get,

$f(x+8) + f(x) = f(x+4)$ _____(ii)

From (i) and (ii), we get,

$f(x+8) + f(x+4) + f(x-4) = f(x+4)$